VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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A PROJECT SYNOPSIS
ON

"AI SMART ATTENDANCE SYSTEM USING FACE RECOGNITION"

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PROJECT SYNOPSIS

AI Smart Attendance System Using Face Recognition

1. Introduction

Recording attendance is one of the most routine yet most important activities of an engineering college. Student attendance decides eligibility for the VTU semester-end examinations, which require a minimum of 75% attendance in every subject. Despite progress in digital campuses, attendance is still taken by calling names, passing a paper register or tapping an RFID card. These methods consume lecture time, depend on the honesty of every student, allow proxy attendance and produce records that must be typed into a portal again.

This project, AI Smart Attendance System Using Face Recognition, replaces these practices with an automated, contact-free and proxy-resistant attendance platform for Rajiv Gandhi Institute of Technology (RGIT), Bengaluru. A student marks attendance from a smartphone or laptop using a teacher-launched session PIN, the device GPS location and a live face capture. Only when every check passes is an attendance record stored.

2. Problem Statement

Existing attendance practices and many simple “face attendance” applications suffer from the following problems:

- Impersonation: photographs, shared cards or a friend answering the roll call defeat the system.
- Location spoofing: check-in applications that do not verify location allow attendance from outside the college.
- No class context: records often show only that a person was seen, not the subject, period or teacher.
- Shared devices: one phone can be used to mark attendance for several students.
- Weak governance: no clear separation between student, teacher, HOD and administrator, and no audit trail.
- Cost: commercial face terminals in every classroom are expensive for most institutions.

The problem addressed is therefore to design and implement a low-cost, role-based attendance platform that identifies the student by face, confirms presence on the campus, ties every record to a teacher-launched class session and keeps an auditable history, using only ordinary phones and laptops.

3. Objectives

1. To automate attendance marking using face recognition so that no roll call or paper register is needed.
2. To prevent proxy attendance through a combination of face verification, GPS geofencing, session PIN and device binding.

3. To link every attendance record to the correct department, semester, section, subject, period and teacher.
4. To provide separate, permission-controlled portals for students, teachers, HODs and the master administrator.
5. To give students and faculty an immediate view of attendance percentage against the VTU 75% requirement.
6. To keep an auditable record of successful and failed attempts, corrections and administrative actions.

4. Scope of the Project

- A web and Android application with four role-specific portals: Student, Teacher, HOD and Master Admin.
- Student self-registration with a VTU USN, personal details and a camera-captured face photograph, followed by HOD approval.
- Face verification using 128-dimensional embeddings with a stored proof snapshot for each attendance.
- GPS geofence verification (Haversine formula, default radius 250 m around the RGIT campus) with accuracy and mock-location guards.
- Seven-period daily timetable, subject catalogues and editable timetables for the nine departments.
- Teacher-launched sessions with a six-digit rolling PIN and dynamic QR code, live roster, manual override and defaulter analytics.
- Device binding, role-based access control with JSON Web Tokens, a security audit log and CSV reports.

The project does not attempt recognition of whole classrooms from CCTV streams and does not integrate with the examination or fee modules of the university; these are future enhancements.

5. Literature Survey

Automated attendance has been studied using RFID, fingerprint, iris, NFC and face recognition. RFID and NFC cards can be transferred between students; fingerprint systems need physical contact; iris systems need specialised hardware. Face recognition is contactless and works with cameras already present in phones and laptops.

Classical approaches include Eigenfaces, Viola–Jones detection, HOG features and Local Binary Patterns. Deep learning methods such as DeepFace, FaceNet, Deep Face Recognition (VGG), ResNet and ArcFace learn compact embeddings in which the distance between two vectors indicates whether they belong to the same person. Eye-blink detection using facial landmarks and texture-based anti-spoofing address presentation attacks. The survey shows that face recognition alone is not enough: location, class context, device identity and governance must also be verified.

6. Proposed System

The proposed system is a decoupled full-stack application organised around four roles. Students register, enrol their face and mark attendance. Teachers launch attendance sessions, monitor the live roster, correct records and export reports. Heads of Department approve registrations and manage faculty, subjects and timetables. The Master Administrator governs all nine B.E. departments, appoints HODs, calibrates the campus geofence and inspects the security audit ledger. Each department keeps its data in its own SQLite database, giving a simple multi-tenant structure.

7. Methodology

The project follows an iterative (spiral) approach. The attendance pipeline consists of the following steps:

1. Enrolment: the student registers with USN, name, e-mail, mobile number and a photograph; the server checks that exactly one face is present.
2. Approval: the HOD approves or rejects the request; on approval the account, student profile and reference photograph are created.
3. Session launch: the subject teacher starts a class session that generates a six-digit PIN and a dynamic QR code.
4. Verification: the student enters the PIN, shares GPS location and shows the face to the camera. The server checks the registered device, campus geofence, session validity and duplicate marking.
5. Face matching: the face is converted to a 128-dimensional embedding and compared with the stored embedding using a Euclidean distance tolerance of 0.55.
6. Recording: on success, the record is saved in the department database with photographic proof, distance from campus centre and match confidence.
7. Reporting: attendance percentage and defaulters are computed against the 75% rule and exported as CSV reports.

8. Technologies Used

Category	Technology
Back-end	Python, FastAPI
Computer vision	OpenCV, dlib, face_recognition library (128-d embeddings)
Database	SQLAlchemy with SQLite (one database per department)
Authentication	PyJWT (JSON Web Tokens), hashlib
Front-end	React 19, Vite
Mobile packaging	Capacitor (Android camera, geolocation, device identity)
Tools	Visual Studio Code, Git, npm

9. Hardware and Software Requirements

Hardware

Component	Minimum	Recommended
Processor	Intel Core i3 (8th Gen) / Ryzen 3	Intel Core i5/i7 / Ryzen 5 or higher
RAM	8 GB	16 GB
Storage	10 GB free	50 GB SSD
Student device	Smartphone with front camera and GPS	Smartphone with 8 MP+ camera and high-accuracy GPS
Network	Campus Wi-Fi / LAN	Gigabit LAN, 5 GHz Wi-Fi

Software

- Operating system: Windows 10/11 (64-bit) for development; Ubuntu 22.04 LTS recommended for production.
- Client: any modern browser (Chrome, Edge, Firefox) or the Android application package.
- All components are open source.

10. Expected Outcomes

- A working attendance platform that removes roll calls and paper registers.
- Significant reduction of proxy attendance through face, location, PIN and device checks.
- Subject-wise, period-wise records linked to the teacher, with instant 75% eligibility information.
- Role-based dashboards for students, teachers, HODs and the administrator.
- An auditable security log and exportable reports, achieved with only ordinary phones and laptops.

11. Future Enhancements

- Deeper anti-spoofing and liveness detection.
- Classroom-level recognition from CCTV streams.
- Integration with the university examination and fee modules.
- Cloud-based deployment for large-scale use and parent/guardian notifications.

12. References

[1] F. Schroff, D. Kalenichenko and J. Philbin, “FaceNet: A Unified Embedding for Face Recognition and Clustering,” CVPR, pp. 815–823, 2015.

[2] J. Deng, J. Guo, N. Xue and S. Zafeiriou, “ArcFace: Additive Angular Margin Loss for Deep Face Recognition,” CVPR, pp. 4690–4699, 2019.

[3] K. He, X. Zhang, S. Ren and J. Sun, “Deep Residual Learning for Image Recognition,” CVPR, pp. 770–778, 2016.

[4] D. E. King, “Dlib-ml: A Machine Learning Toolkit,” Journal of Machine Learning Research, vol. 10, pp. 1755–1758, 2009.

[5] T. Soukupova and J. Cech, “Real-Time Eye Blink Detection Using Facial Landmarks,” CVWW, 2016.

[6] N. Dalal and B. Triggs, “Histograms of Oriented Gradients for Human Detection,” CVPR, vol. 1, pp. 886–893, 2005.

[7] P. Viola and M. Jones, “Rapid Object Detection Using a Boosted Cascade of Simple Features,” CVPR, vol. 1, pp. 511–518, 2001.

[8] J. W. Yap and M. S. Jawad, “Real Time Face Recognition Attendance System Using Python and OpenCV,” Applied Information Technology and Computer Science, vol. 5, no. 1, pp. 1310–1329, 2024.

[9] M. Jones, J. Bradley and N. Sakimura, “JSON Web Token (JWT),” RFC 7519, IETF, 2015.